OpenGeoSys Training 2026 – KIGAM

Lecture 1: Continuum Mechanics

Olaf Kolditz, Mehran Ghasabeh

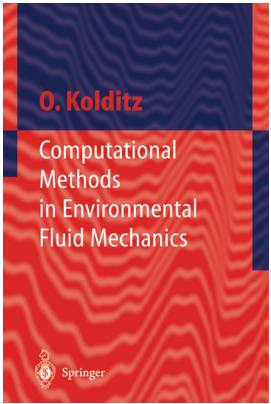
[1]Helmholtz Centre for Environmental Research – UFZ, Leipzig
[2]Technische Universität Bergakademie Freiberg – TUBAF, Freiberg
[3]Technische Universität Dresden – TUD, Dresden

Daejeon, 11th October 2026

DECOVALEX 2027

Agenda

1. OGS Course Overview
2. Contiuum Mechanics
   - OGS-THMC
   - Porous Media Theory
   - Lagrange and Euler Perspectives
   - General Balance Equation
   - Fluxes
   - Conservation Quantities
   - List of Symbols
3. References

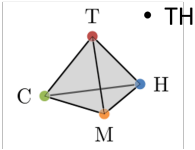


Lectures

- Continuum Mechanics
- Hydromechanics
- Geomechanics

OGS Framework

- ogs^{THMC}
- ogs^{Tools}
- ogs^{Workflows}
- ogs^{DigitalTwins}



Exercises

- ogs installation
- H: Theis: Pumping test
- H: STIMTEC: Hydraulic tests
- H: Selke: Catchment
- M: Kirsch: Excavation
- HM: Fault reactivation
- THM: FE experiment

Part 1 morning session
lectures and software demonstration

Part 2 afternoon session (optional)
hands-on work with OGS (exercises)

Olaf

Lectures

- L0 Course Overview | OpenGeoSys Introduction
- L1 Continuum Mechanics
- L2 Hydromechanics
- L3 Groundwater Modeling

Overleaf: `https://www.overleaf.com/read/gpfdtpzqrjvp#ffc4c4`

Exercises

- E0 OpenGeoSys: OGS basics, Benchmark Gallery, GitHub
- E1 Theis: Pumping well
- E2 STIMTEC: Pumping test in fractured rock
- E3 Selke: Catchment modeling

GitHub: `https://github.com/OlafKolditz/ogs-course-2026`

DECOVALEX 2027

Mehran

Lectures

- L4 Underground Tunnel Excavation
- L6 Hydro-Mechanically Coupled Processes in Fault-Controlled Porous Medium
- L7 Heated Tunnel Problem

Exercises

- E0 Tunnel Convergence Problem
- E1 Theis: Pumping well
- E2 STIMTEC: Pumping test in fractured rock
- E3 Selke: Catchment modeling

github:
`https://github.com/mehranqsb/OGSTrainingCourse_KIGAM_ZJU`

Lecture 1

Continuum Mechanics

OpenGeoSys THMC



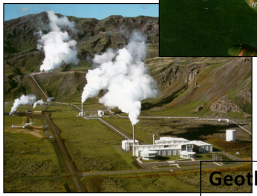
Water Resources





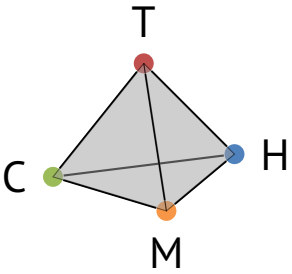
Water Quality

Geotechnical Systems



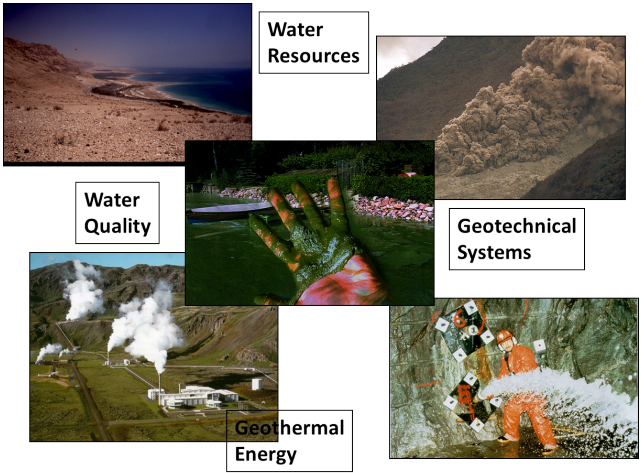


Geothermal Energy



DECOVALEX 2027

OpenGeoSys THMC



ogs-applications:

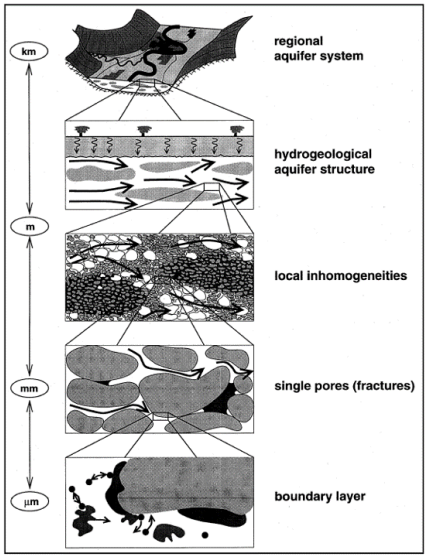
- T geothermal
- H geohydrological
- M geomechanical
- C geochemical (RTP)

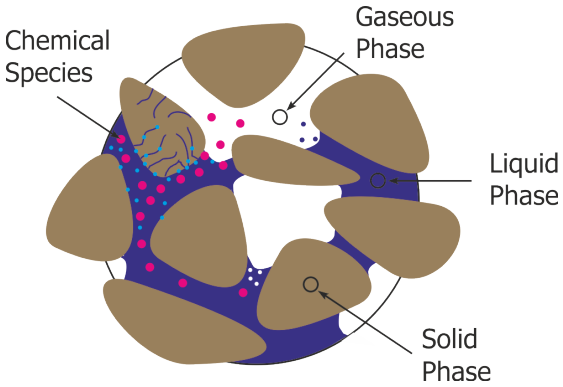
ogs-brands:

- ogs^{THMC}
- ogs^{Tools}
- ogs^{Workflows}
- ogs^{DigitalTwins}

DECOVALEX 2027

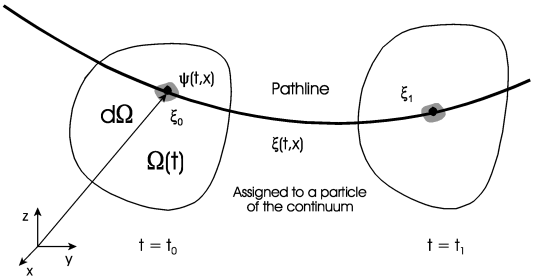
Porous Media Theory



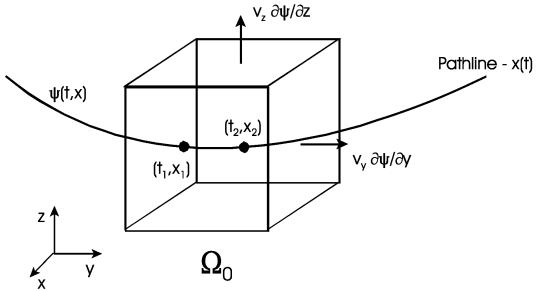


- multi-component: $\sum_k$
- multi-phase: $\sum_\gamma$

Lagrange concept



Euler concept



$$\frac{d}{dt}\int_{\Omega}\psi d\Omega = \int_{\Omega_0}\frac{\partial\psi}{\partial t}d\Omega + \oint_{\partial\Omega_0}\psi(t)\mathbf{v}\cdot d\mathbf{S} = \int_{\Omega} q^{\psi} d\Omega \qquad (1)$$

Reynolds Transport Theorem: Euler - Lagrange Equivalence

Gauss-Ostrogradski Theorem

$$\oint_{\partial\Omega_0} \Phi^{\psi} \cdot d\mathbf{S} = \int_{\Omega} \nabla \cdot \Phi^{\psi} d\Omega_0 \tag{2}$$

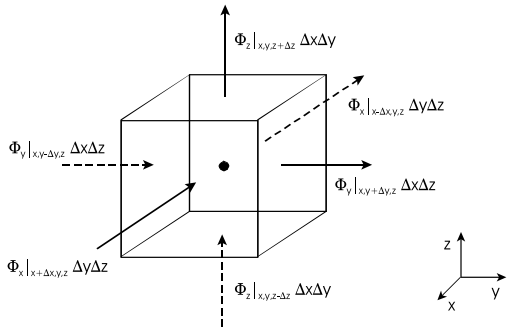
General Balance Equation of Continuum Mechanics

$$\frac{d}{dt}\int_{\Omega} \psi d\Omega = \int_{\Omega} \frac{\partial \psi}{\partial t} d\Omega_0 + \int_{\Omega} \nabla \cdot \Phi^{\psi} d\Omega_0 = \int_{\Omega} q^{\psi} d\Omega \tag{3}$$

$$\forall \Omega : \frac{d\psi}{dt} = \frac{\partial \psi}{\partial t} + \nabla \cdot \Phi^{\psi} = q^{\psi} \tag{4}$$

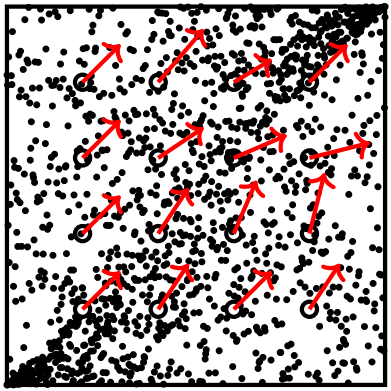
The total flux Φ^{ψ} of a quantity ψ is defined as

$$\Phi^{\psi} = \mathbf{v}^{E}\psi \tag{5}$$



$$\mathbf{v}^{E} = \cup\,\mathbf{v}_i = \bar{\mathbf{v}} + \hat{\mathbf{v}}$$

$$\hat{\mathbf{v}} = \mathbf{v}^{E} - \bar{\mathbf{v}}$$



$$\Phi^{\psi} = \mathbf{v}^{F}\psi = \underbrace{\bar{\mathbf{v}}\psi}_{\Phi_A^{\psi}} + \underbrace{(\mathbf{v}^{F} - \bar{\mathbf{v}})\psi}_{\Phi_D^{\psi}} \tag{6}$$

decomposed into two parts:
advective flux Φ_A^{ψ} and diffusive flux Φ_D^{ψ} relative to the mass-weighted velocity:

- advective flux of quantity ψ

$$\Phi_A^{\psi} = \mathbf{v}\psi \tag{7}$$

- diffusive flux of quantity ψ (Fick's law)

$$\Phi_D^{\psi} = -\mathbf{D}\cdot\nabla_ \tag{8}$$

$$\bar{\mathbf{v}} \equiv \mathbf{v}$$

- Integral form

$$\int_{\Omega} \frac{d\psi}{dt} d\Omega = \int_{\Omega_0} \frac{\partial \psi}{\partial t} d\Omega + \int_{\Omega_0} \nabla \cdot (\mathbf{v}\psi) d\Omega - \int_{\Omega_0} \nabla \cdot (\mathbf{D}^{\psi} \nabla \psi) d\Omega = \int_{\Omega} Q^{\psi} d\Omega \tag{9}$$

- Differential form

$$\frac{d\psi}{dt} = \frac{\partial \psi}{\partial t} + \nabla \cdot (\mathbf{v}\psi) - \nabla \cdot (\mathbf{D}^{\psi} \nabla \psi) = Q^{\psi} \tag{10}$$

The amount of a quantity in a defined volume Ω is given by

$$\Psi = \int_{\Omega} \psi d\Omega(t) \tag{11}$$

where Ψ is an extensive conservation quantity (i.e. mass, momentum, energy) and ψ is the corresponding intensive conservation quantity such as mass density ρ, momentum density $\rho\mathbf{v}$ or energy density e.

Extensive quantity Ψ	Symbol	Intensive quantity ψ	Symbol
Mass	M	Mass density	ρ
Linear momentum	$\mathbf{m}$	Linear momentum density	$\rho\mathbf{v}$
Energy	E	Energy density	$e = \rho i + \frac{1}{2}\rho v^2$

List of Symbols

Ω	control volume / spatial domain
Ω_0	reference (material) domain
$\partial\Omega_0$	boundary of the reference domain
t	time
$\frac{d}{dt}$, $\frac{d\psi}{dt}$	material (total) time derivative
$\frac{\partial\psi}{\partial t}$	partial (local) time derivative
$\int_\Omega$	volume integral
$\oint_{\partial\Omega_0}$	closed surface integral over the boundary
$d\Omega$, $d\Omega_0$	volume element
$d\mathbf{S}$	vector surface element
∇	nabla (gradient) operator
$\nabla\cdot$	divergence operator
$\forall$	“for all”
$\equiv$	“is identical to / defined as”
$\sum_k$	summation over components k
$\sum_\gamma$	summation over phases γ
$\cup$	union
$\mathbf{v}$	velocity vector
$\mathbf{v}^E$	barycentric / effective (mixture) velocity
$\mathbf{v}_i$	velocity of constituent i
$\bar{\mathbf{v}}$	mass-averaged (mean) velocity
$\hat{\mathbf{v}}$	diffusion velocity

ψ	generic intensive (specific) conservation quantity
Ψ	corresponding extensive conservation quantity
Φ^ψ	total flux of quantity ψ
Φ^ψ_A	advective flux
Φ^ψ_D	diffusive flux
$\mathbf{D}^\psi$	diffusion coefficient/tensor for ψ
q^ψ	specific source/supply term
Q^ψ	volumetric source term (differential form)
M	mass
$\mathbf{m}$	linear momentum
E	energy
ρ	mass density
$\rho\mathbf{v}$	linear momentum density
e	energy density, $e = \rho i + \frac{1}{2}\rho v^2$
i	specific internal energy
v^2	squared velocity magnitude

References

[1] J. Bear. Modeling Phenomena of Flow and Transport in Porous Media. Theory and Applications of Transport in Porous Media. Springer International Publishing, 2018. ISBN: 9783319728261. URL: https://books.google.de/books?id=KxxJDwAAQBAJ.

[2] Jacob Bear. Dynamics of Fluids in Porous Media. Environmental Science Series. New York: American Elsevier Publishing Company, 1972. ISBN: 9780444001146. URL: https://books.google.com/books?id=9JVRAAAAMAAJ.

[3] Olivier Coussy. Poromechanics. 2nd. Chichester, England: John Wiley & Sons, Ltd, 2004. ISBN: 9780470849200. DOI: 10.1002/0470092718.

[4] Hans-Jörg G. Diersch. FEFLOW: Finite Element Modeling of Flow, Mass and Heat Transport in Porous and Fractured Media. Springer Berlin, Heidelberg, 2014. ISBN: 978-3-642-38739-5. DOI: 10.1007/978-3-642-38739-5.

[5] Wolfgang Ehlers and Joachim Bluhm, eds. Porous Media: Theory, Experiments and Numerical Applications. Springer Berlin, Heidelberg, 2002. ISBN: 978-3-662-04999-0. DOI: 10.1007/978-3-662-04999-0.

[6] Olaf Kolditz. Computational Methods in Environmental Fluid Mechanics. Springer Berlin Heidelberg, 2002. ISBN: 9783662047613. DOI: 10.1007/978-3-662-04761-3. URL: http://dx.doi.org/10.1007/978-3-662-04761-3.

DECOVALEX 2027